

CSA1706-ARTIFICIAL INTELLIGENCE

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1. Water Jug Problem (4-Gallon and 3-Gallon Jugs) Problem

There are two jugs:

- Jug A = **4 gallons**
- Jug B = **3 gallons**

Goal:

- Obtain **exactly 2 gallons in the 4-gallon jug.**

State Representation

State = (A, B)

Where:

- A = Water in 4-gallon jug
- B = Water in 3-gallon jug

Initial State:

(0,0)

Goal State:

(2,x)

where x can be 0–3.

Solution Steps

Step	Action	State (4G,3G)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon jug until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

Goal Achieved

4-Gallon Jug = 2 Gallons

State Space Diagram

(0,0)

|

Fill 3G

|

(0,3)

|

Pour to 4G

|

(3,0)

|

Fill 3G

|

(3,3)

|

Pour to 4G

|

(4,2)

|

Empty 4G

|

(0,2)

|

Pour to 4G

|

(2,0) Goal

2. Mars Rover Agent

i) Percepts

The rover receives information from:

- Cameras
 - Navigation sensors
 - Temperature sensor
 - Rock detector
 - Soil analyzer
 - Spectrometer
 - GPS/Localization sensors
 - Obstacle sensors
 - Battery level
 - Communication signals
-

ii) Operating Environment

Property	Description
Observable	Partially Observable

Property	Description
Agents	Single Agent
Deterministic	Mostly Stochastic
Sequential	Yes
Dynamic	Yes
Continuous	Yes
Known	Partially Known

iii) Actions

The rover can:

- Move Forward
- Move Backward
- Turn Left
- Turn Right
- Stop
- Capture Images
- Drill Rocks
- Collect Soil
- Analyze Samples

- Avoid Obstacles
 - Send Data to Earth
 - Recharge using Solar Panels
-

iv) Performance Measure

Performance is evaluated by:

- Number of samples collected
 - Accuracy of analysis
 - Distance travelled safely
 - Energy efficiency
 - Number of obstacles avoided
 - Successful communication with Earth
 - Mission completion time
-

v) Suitable Agent Architecture

Goal-Based + Utility-Based Agent

Reason

Because the rover must

- Reach target locations
- Avoid obstacles

- Save battery power
- Select best routes
- Maximize scientific discoveries

Utility helps choose the **best action** among many alternatives.

3. 8-Queens Problem

Problem Formulation

Initial State

Empty Chessboard

State

Arrangement of queens already placed.

Example

Q

.. Q

. . . . Q . .

Actions

Place one queen in the next column where no attack occurs.

Goal Test

All 8 queens are placed without attacking each other.

Conditions:

- No same row
 - No same column
 - No same diagonal
-

Path Cost

Every move costs 1.

Total cost = 8

Search Space

Empty Board

|

Place Queen Column 1

|

Possible Rows

|

Place Queen Column 2

|

Safe Positions

|

...

|

Column 8

|

Goal State

Search Strategy

Typically solved using:

- Backtracking Search
- Depth First Search (DFS)

Backtracking removes incorrect queen placements and tries another safe position.

4. OLA Cab Booking Problem

Type of Problem-Solving Agent

Utility-Based Agent

Reason:

The customer chooses the best cab based on:

- Fare
- Comfort
- Waiting Time

- Distance
- Availability
- Travel Time

The agent selects the option with the highest utility.

Pseudo Code

START

Input Pickup Location

Input Destination

Read Available Cabs

FOR each Cab

 Calculate Fare

 Calculate Waiting Time

Check Availability

Calculate Utility

END FOR

Choose Cab with Maximum Utility

Display Cab Details

Book Cab

END

Flow

Customer Request

|

Find Nearby Cabs

|

Calculate Fare

|

Compare Cab Types

|

Select Best Cab

|

Confirm Booking

5. Uniform Cost Search (UCS)

Algorithm

Uniform Cost Search expands the node having the **minimum cumulative path cost**.

Pseudo Code

UCS(Start, Goal)

Create Priority Queue

Insert Start with Cost = 0

WHILE Queue is not Empty

 Remove Node with Lowest Cost

 IF Goal Found

 Return Path

 Expand Current Node

FOR each Neighbor

New Cost = Current Cost + Edge Cost

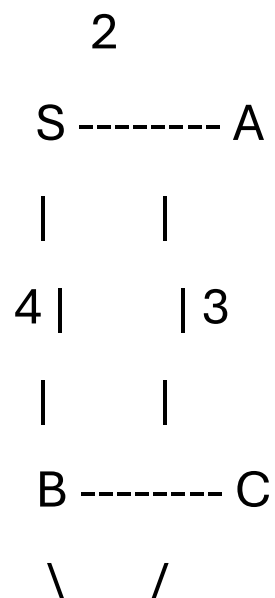
IF Better Cost

Insert into Queue

END WHILE

Failure

Example Weighted Graph



5\ /2

\ /

G

Working

Priority Queue

Start

S (0)

Expand S

A (2)

B (4)

Expand A

C (5)

Expand B

G (9)

Expand C

G (7)

Expand G

Goal Found

Least-Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Cost

$S \rightarrow A = 2$

$A \rightarrow C = 3$

$C \rightarrow G = 2$

Total Cost = 7

Summary

Question Answer

- 1 Water Jug Problem solved with state-space search; goal state **(2,0)**
- 2 Mars Rover: Percepts, environment, actions, performance, and **Goal/Utility-Based Agent**
- 3 8-Queens formulated with **initial state, actions, goal test, path cost**, solved using **Backtracking/DFS**
- 4 OLA Cab Booking uses a **Utility-Based Agent** with pseudocode to select the best cab
- 5 Uniform Cost Search expands the lowest-cost path first; example solution **S → A → C → G** with **total cost = 7**